Discover Plants

Review

Neem (*Azadirachta indica* A. Juss): a multifaceted tree & an elixir in the traditional system of Indian medicine

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Abstract
Neem, called '*Arista*' in Sanskrit, meaning 'reliever of sickness', often referred to as 'the bitter gem', is one of the most celebrated plants in the Indian Traditional Healthcare system for its diverse medicinal properties. Scientifically, *Azadirachta indica*, is classified in the Mahogany (**Meliaceae**) family. Neem stands out as a beacon of promise for the twenty-first century. Each part of the Neem tree, i.e., its flowers, leaves, fruits, seed, seed oil, Neem cake, and gum, has therapeutic properties. Researchers have identified nearly 300 distinct phytochemicals within the Neem tree, having diverse pharmacological effects like antimicrobial, antiviral, antifungal, anti-inflammatory, anti-plasmodial, antiseptic, antipyretic, anti-diabetic, and anti-ulcer activities thereby justifying the symbolic term 'elixir' written in the title of this review. Neem is mentioned in ancient Ayurvedic treatises like *Charaka Samhita*, *Susruta Samhita*, *Ashtanga Hridaya* (a book authored by *Vaghbata*) and in *nighantus* (medical lexicon) written over the ages. In Ayurveda, plenty of formulations contain Neem as an ingredient. This review sheds light on *Azadirachta indica* A. Juss (Neem plant) from the standpoint of the traditional system of Indian medicine *vis-à-vis* its taxonomical position, macroscopic and organoleptic characteristics, bio-activities, use as bio-pesticide and bio-fertilizer, active phytoconstituents, and pharmacological studies. Additionally, it presents a compilation of folklore recipes and Ayurvedic, Homeopathic, Siddha and Unani formulations (ASU & H) that contain Neem as an ingredient.

Keywords *Azadirachta indica* (Neem) · Ayurveda · Bio-actives · Phytochemicals · Pharmacology · Ayurvedic · Homeopathic · Siddha and Unani formulations

Abbreviations

Enm.	Endosperm
Fl.	Flower
Fr.	Fruit
Lf.	Leaf
Sd.	Seed
St. Bk.	Stem bark
Rt. Bk.	Root bark

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Fl. Bd. Flower bud
WP Whole plant

1 Introduction

Medicinal plants are one of the nature's invaluable gifts and have been used from the remote past to treat various human diseases across the globe. In rural areas of underdeveloped and developing countries, approximately 80% of people rely on plant-based extracts, i.e., traditional medicine, which serves as the primary source of healthcare needs [1]. One of the key issues with modern drugs is their side effects, which has significantly contributed to a growing interest in traditional medicine like Ayurveda, and a surge in the use of herbal products in recent years has also been observed [2]. Plants also provide essential nutrients, including macronutrients (carbohydrates, proteins, and fats), micronutrients (minerals and vitamins) and secondary metabolites (alkaloids, phenolics, tannins, flavonoids, etc.), all of which have beneficial effects on human health [3].

The Neem (*Azadirachta indica*) is one of the most celebrated, multifaceted and versatile medicinal plants known for its numerous health benefits. Tons of benefits and therapeutic properties associated with the Neem plant have been documented in hymns (*shloka*) and texts from time immemorial [4]. The *shloka* (Sanskrit verse) (see footnote) depicting the health benefits of the Neem (*A. indica*) plant is just an indicative one.[1] The mention of *A. indica* (Sanskrit: *Nimba* or *Picumaradah*, commonly called Neem or Indian lilac) can be found in ancient Ayurvedic texts comprising *Charaka Samhita*, *Susruta Samhita* (written during 1500 B.C.–400 A.D. by *Acharya Charaka* and *Acharya Susruta* respectively), book (*Ashtanga Hridaya*) authored by *Vaghbata* (~ 500 AD) as well as in *nighantus* (medical lexicon) written across the ages. Trans-Disciplinary University, Bengaluru's bibliographic database highlights that '*Nimba*' is extensively documented in Ayurvedic scriptures, with 448 mentions across 20 foundational texts dated between 1500 BC and 1900 AD [5]. In Sanskrit, the Neem tree is called '*Arista*', meaning 'reliever of sickness', or '*sarba-rogaribarini*', which means it has the potency to alleviate diseases [6]. Its depiction is found not only in *Nighantus* (*Dhanvantari Nighantu* [7], *Bhavprakash Nighantu* [8], *Priya Nighantu*, *Kaidev Nighantu* [9], *Dravyaguna Nighantu* [10]) but also in Vedic texts (*Atharva Veda*) [11], epics and Ayurvedic Pharmacopoeia of India (API) [12]. For example, the *Ramayana*, dating back to the *Tretayuga* around 3000 BC, lists *Nimba* as one of the significant trees in *Nandavana*. The neem tree, as it is commonly known, is celebrated in various classical works, including the *Brhat Samhita* and *Padma Purana*. Astrologically, Neem is recommended as a protective measure for those born under the star *Purvabhadra*. Ayurveda attributes over 50 different Sanskrit names to the Neem tree, reflecting its diverse morphological and therapeutic characteristics [5]. It is called '*Kakaphala*' because crows consume its fruits. Neem is recognised for inducing vomiting, yet it is also an antiemetic, earning the name '*Chardighna*'. It's called '*Hingu Niryasa*' due to the gum it secretes, resembling asafetida. Its anti-inflammatory abilities are captured in the name '*Puyari*', with '*puya*' meaning pus and '*ari*' meaning enemy. 'Krimighna' underscores its effectiveness against worms and infections, while '*Picumarda*' relates to its skin-healing powers [13]. The *Charaka Samhita* details Neem's applications for skin conditions, diabetes, and more. Interestingly, *Charaka's* emphasis on Neem's dermatological and metabolic benefits differs from that of other scholars, such as *Susruta*, who prioritises its anti-inflammatory effects. *Susruta* also endorses Neem as the optimal choice for oral hygiene, particularly for brushing teeth, and as a treatment for *nadivrana*, a type of ulcerative vascular disease [14]. The Neem tree is well known for providing an extensive range of non-timber products, including its leaves, bark, flowers, fruit, seeds, gum, oil, and Neem cake. The economic and ecological benefits are significant because the tree does not need to be felled to harvest these valuable resources. Apart from the traditional medicine system, in the modern era, Neem is celebrated for its diverse medicinal properties, such as anti-allergenic, anti-inflammatory, antifeedant, and anti-fungal, and acts as an antiseptic, scabicide, heart tonic, diuretic, insecticide, larvicide, nematicide, and spermicide etc. [15, 16]. These attributes have made Neem a highly sought-after plant in various applications [17]. Besides, Neem oil is used as a holistic pesticide and has widespread applications in crop protection [18]. In addition, in recent years, the incorporation of Neem or Neem derived products into skincare and cosmetics has

[1] शतायुर्वज्रदेहाय सर्व सम्पत्कराय च ।
सर्वारिष्ट-विनाशाय निम्बकं दल-भक्षणम् ॥
śatāyurvajradēhāya sarva sampatkarāya ca ।
sarvāriṣṭa-vināśāya nimbakaṁ dalabhakṣaṇam ॥
For longevity of 100 years, for a body as strong as a diamond, for all prosperity and wealth, for the eradication of all calamities, eat the tender leaves of Neem.

surged due to a rising demand for natural and sustainable ingredients. Beyond its cultural significance, Neem's versatility has positioned it at the forefront of the beauty industry. Its role spans dermatology and beauty rituals, providing a comprehensive understanding of Neem's diverse contributions to skincare and cosmetics [19, 20]. Neem (methanolic leaf extract) has also been utilized in the in vivo production of stem cells [21].

2 Taxonomical position of Neem plant

The scientific name of the Neem tree is *Azadirachta indica*. The term "*Azadirachta*" originates from the Persian word "azaddhirakt," which means "Noble tree." Neem plant comes under the Mahogany family. The detailed taxonomy of the Neem plant is listed in the following table (Table 1).

3 Macroscopic and organoleptic characteristics of Neem

The macroscopic and organoleptic properties of various Neem parts are listed below. Table 2 documents the macroscopic and organoleptic characteristics of Neem.

4 Neem tree at a glance

The evergreen species Neem tree can grow to medium or large sizes up to 15 to 20 m or even taller when conditions are favourable. It grows across the flatlands of the Indian subcontinent up to an elevation of 900 m [12, 27]. It is also found in Southeast Asian countries, the African continent, the Caribbean islands, and Central and South American countries [28]. Neem trees can survive well in dry conditions and are cultivated to combat soil erosion and promote conservation. For the proper growth of the Neem tree, pH is required to be between 4 and 10. It enhances the soil's nutrient content and water retention capacity, can neutralise acidic soils, and is beneficial in reviving deteriorated lands [29, 30].

In Ayurveda, the Neem tree is utilised for skin care and enhances the skin's beauty [31]. The Neem tree is one of the most multifaceted, versatile and potent plants grown in tropical regions. It offers the most significant number of beneficial non-timber products, which includes leaves, bark, flowers, fruit, seeds, gum, oil, and Neem cake, more than any other tree species, which is beneficial as well as economical from the angle is that the whole tree need not be cut for acquiring those products. It is recognised for a wide array of medicinal properties, i.e., anti-allergenic, anti-inflammatory, antifeedant, and anti-fungal, and acts as an antiseptic, scabicide, heart tonic, diuretic, insecticide, larvicide, nematicide, and spermicide. These properties have led to widespread use, establishing Neem as a valuable ecological asset [32, 33]. Figures 1 and 2 document photographs of various parts of the Neem and the various applications, activities and uses of Neem, respectively.

Table 1 Taxonomical classification of *A. indica* [22]

Domain:	Eukaryotae
Kingdom:	Plantae
Phylum:	Spermatophyta
Sub-phylum:	Angiosspermae
Class:	Dicotyledonae
Order:	Rutales
Suborder:	Rutinae
Family:	Meliaceae (Mahogany family)
Subfamily:	Melioideae
Tribe:	Melieae
Genus:	*Azadirachta*
Species:	*Indica*

Table 2 Macroscopic and organoleptic characteristics of Neem tree [23–26]

Parts	Stem bark (St. Bk.)	Leaf (Lf.)	Flower (Fl.)	Fruit (Fr.)	Seed (Sd.)	Seed oil
Organoleptic properties						
Colour	Rusty grey (external); yellowish (internal)	Yellowish green	Pale yellow, white	Green (initially); Yellowish brown, chocolate (mature)	Brown (external); creamish (internal)	Yellow to brown
Odour	Characteristic	Indistinct	Honey scented	Characteristic	Aromatic/garlic (when crushed)	Garlic
Taste	Bitter	Bitter	Bitter	Sweet to semi-sweet pulp	Bitter	Bitter
Size	Little bit curved to flat, but it varies as per the age of the tree	Length: 7–8.5 cm; Width: 1–1.7 cm	Length: 5–6 cm; Width: 8–11 cm	Length: 0.5–2 cm; Width: 0.5–1.2 cm; Thickness: 0.6–0.9 cm	Length: 10–15 mm; Width: 3–5 mm; Thickness: 3–4 mm	–
Shape	Rough surface (external); foliaceous (internal)	Lanceolate, serrate, acute	Drooping panicles	Smooth surface; oval to nearly round-shaped	Irregular	–

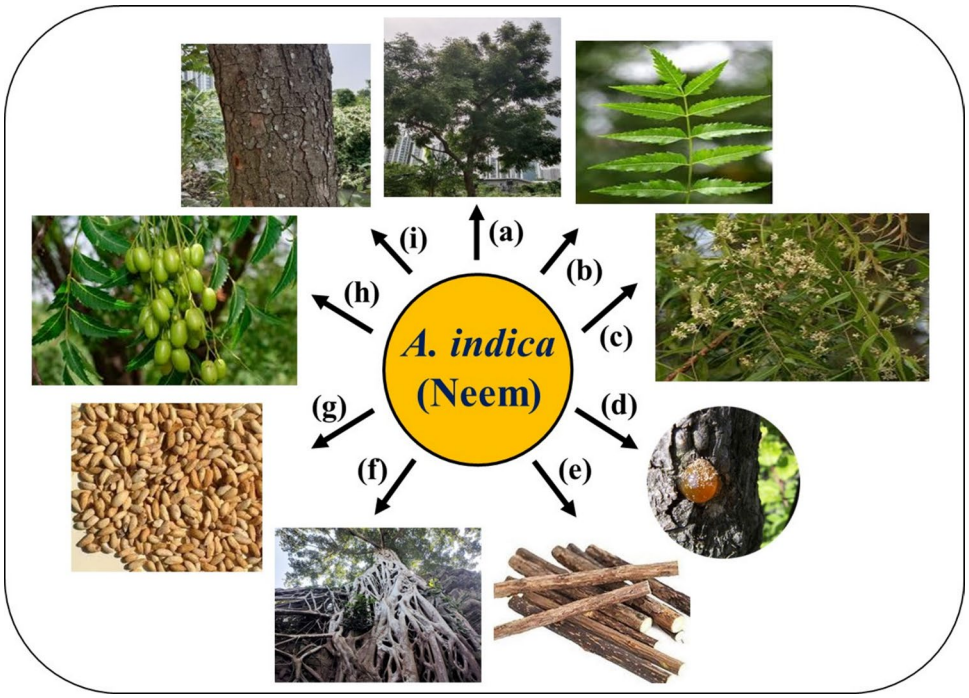


Fig. 1 Various parts of Neem tree: **a** Whole plant, **b** Leaf, **c** Flower, **d** Gum, **e** Twig, **f** Root, **g** Seed, **h** fruit and **i** Stem bark

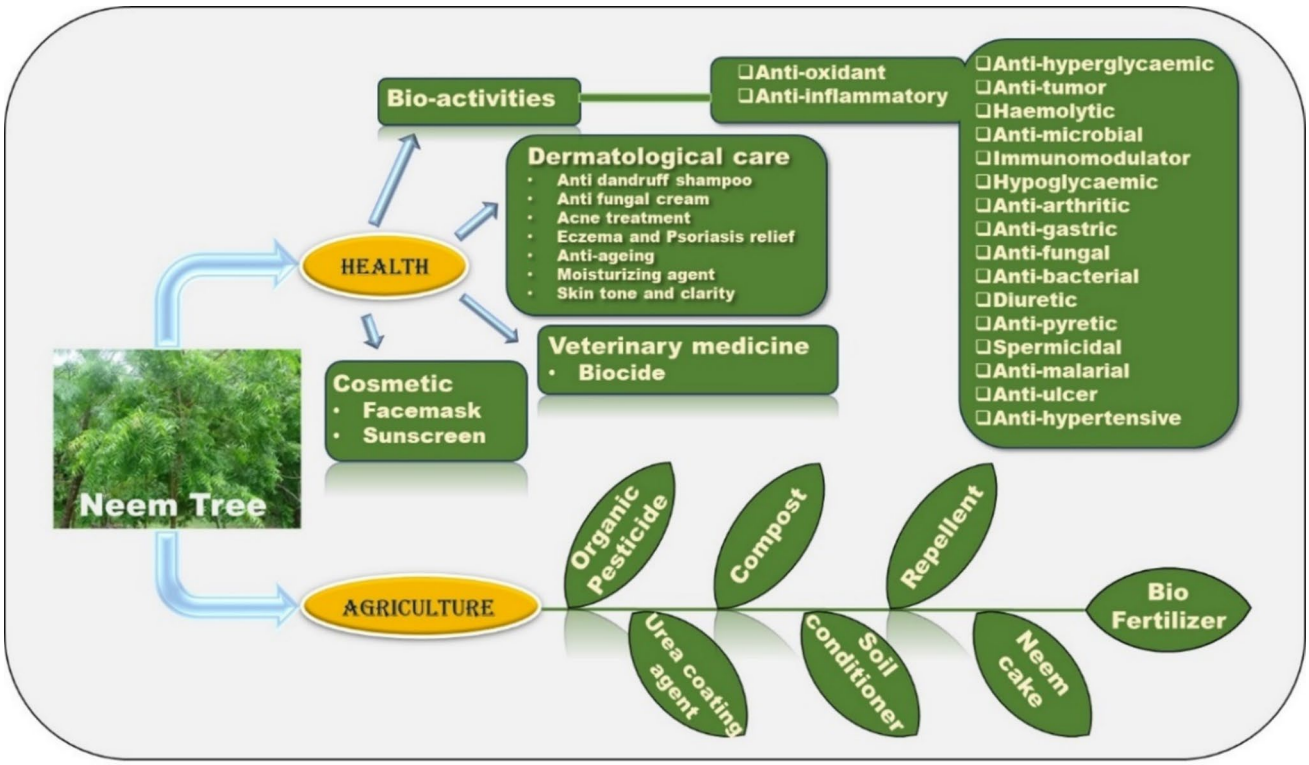


Fig. 2 Various applications, activities and uses of Neem

Table 3 Synonyms of *A. indica* in different languages [12]

Sl. No	Language	Synonyms	Sl. No	Language	Synonyms
Scientific Name: *A. indica*					
01	Assamese:	Mahanim	12	Oriya:	Neemo, Nimba
02	Bengali:	Nim, Nimgaachh	13	Punjabi:	Nimb, Nim
03	Gujarati:	Leemado	14	Sanskrit:	Picumaradah, Aristah, Picumandah, Prabhadrah
04	Hindi:	Neem	15	Sindhi:	Neem
05	Kannada:	Turakbevu, Huchchabevu, Chikkabevu	16	Tamil:	Vempu, Veppu
06	Kashmiri:	Neem	17	Telugu:	Vemu, Vepa
07	Konkani:	Kadulimb	18	Urdu:	Neem
08	Malayalam:	Veppu, Aryaveppu, Aaruveppu	19	Bodo:	Nim
09	Manipuri:	Neem	20	Santali:	Nim
10	Marathi:	Kadunimba, Nimb	21	Maithili:	Neem
11	Nepali:	Neem	22	Dogri:	Neem

Apart from the above information, known out of research, our ancient texts also mention the therapeutic effects of the Neem tree. The *shloka* (Sanskrit verse) cited in the footnote describes the therapeutic effects of the Neem tree and its various parts in detail [34], an English translation of the *shloka* is given for easy understanding.[2]

India is a diverse country with nearly 1600 languages spoken. The 8th schedule of the Indian Constitution has given the official language status to only (22) languages. The following table (Table 3) documents the synonyms of *A. indica* in those official languages.

[2] निम्बस्तिक्तः कटुः पाके लघुः शीतोऽग्निवातकृत् ।
ग्राह्यहृद्यो जयेत् पित्तकफमेहज्वरकृमीन् ।
कुष्ठकासारुचिश्वासहृल्लासश्वयथुव्रणान् ।
ग्राहि प्रवाळं निम्बस्य रक्तपित्तकफकृमीन् ।
कुष्ठघ्नं वातजननं नेत्ररोगान् विनाशयेत् ।
तद्वत् पत्रणि निम्बस्य व्रणघ्नानि विशेषतः ।॥
शलाका निम्बपत्रस्य कासश्वासविनाशिनी ।
कृमिघ्ना तु वरिष्ठा स्यात् कुष्ठज्वरविनाशिनी ॥
चक्षुष्यं निम्बपुष्पं च कृमिपित्तविषप्रणुत् ।
वातळं कटुपाकं स्यात् सर्वारोचकनाशनम् ।
फलं तिक्तं रसे पाके कटुकं भेदनं लघु ।
अरूक्षमुष्णं कुष्ठघ्नं गुल्मार्शःकृमिमेहनुत् ।
निम्बस्य पक्वं मधुरं सतिक्तं.
स्निग्धं फलं शोणितपित्तरोगे ।
कफे प्रशस्तं नयनामयघ्नं.
क्षतक्षयघ्नं गुरु पिच्छिलं च ॥
निम्बबीजस्य मज्जा च कृमिकुष्ठविशोधनः ।
नात्युष्णं निम्बजं तिक्तं कृमिकुष्ठकफप्रणुत् ॥
अभ्यङ्गान्नावनात् क्षीरभोजिनः पलितापहम् । (कै.नि.).

English translation of the above hymn: *Neem is bitter, light in property, cold in potency, pungent post digestion, increases digestion and causes vitiation of vata dosha, is water absorbent, cordial to the heart, cures pitta and kapha dosha, diabetes, fever, worm infestation, skin disorders, cough, taste lessness, dyspnoea, nausea, oedema and wounds. Tender neem leaves are water absorbent and cause vitiation of vata dosha and kapha dosha, hemorrhagic disorders, worm infestation, skin disorders and eye disorders, whereas dry Neem leaves alleviate wounds. Neem flowers are pungent post digestion, beneficial to the eyes, cause vitiation of vata, kapha and pitta dosha, poisoning, and all types of taste disorders. Neem fruit is bitter in taste, hot in potency, light in property, pungent post digestion, piercing in nature, cures skin disorders, abdominal tumours, piles, worm-related ailments, and diabetes. Neem ripe fruits are sweet and bitter, unctuous, heavy, and slimy in quality, and cure hemorrhagic disorders, vitiation of kapha dosha, eye disorders, wounds, and weakness.*

5 Folklore about Neem

In the Indian context and as per Hindu mythology, the Neem tree is symbolised by the *Goddess Durga*, the wife of *Lord Shiva*, commonly known as *Devi Parvati*. In some regions of India, the Neem tree is considered a divine entity. *Neemari Devi*, whose other name is *Goddess Sitala Devi* in the north, as well as *Goddess Marimman* in the south, are both known to heal skin ailments like smallpox. In Odisha, the famous Jagannatha deities are made up of Neem wood. In traditional healing practices, Neem leaves and twigs are utilised in tantra and mantra therapies to protect against evil spirits. The Neem tree is believed to transmit the healing potential from the physician/exorcist to the victim. In Ayurveda, *Bhoot Vidya* (Psychiatry) refers to the practice of exorcising evil entities, which falls under the category of Mantra therapy, which includes *Mani* (Gem), *Mantra* (Spells), and *Ousadham* (Medicine). Specifically, Neem leaves are employed in mantra therapy to treat venomous bites from snakes and scorpions. This is done by reciting holy mantras and performing tantric ceremonies, where traditional healers use Neem twigs with leaves on the affected person. Apart from this, Neem flowers and leaves are used in Indian cooking. [35, 36].

6 Recipes with Neem

Although Neem is bitter, considering its numerous health benefits, in Indian tradition, it is used in recipes, mainly leaves and flowers. *Neem Aloo Vaja* (Neem Potato Fry), *Neem Baigan Sabji* (Neem Brinjal Curry), Neem juice, Neem tea, *Neem chutney* (Neem Sauce), *Vepampoo Rasam* (Neem Flower Broth), Neem Flower cutlet, *Neem Narkel Bhaja* (Neem Coconut Fry), *Neem Leaf Pakoda* (Neem Leaf Fritter), Neem Sukto (Bitter mixed vegetable stew), *Neem Bori* (Dried lentil nuggets made with Neem), *Neem Papad* (Neem crispbread) (Fig. 3) are a few preparations made from Neem.

7 Neem as an organic pesticide

In the 1990s, the agricultural sector faced pest and disease control challenges due to new legislation like the Food Quality Protection Act and Worker Protection Standard, which restricted traditional chemical pesticides. This led to a shift towards integrated pest management (IPM) and using natural motifs. Neem-based insecticides, notably those containing azadirachtin, emerged as a solution. Azadirachtin is effective against over 300 insect species, including significant pests



Fig. 3 Various recipes prepared with Neem

Fig. 4 Neem cake



Table 4 Major nutrient elements present in Neem cake [42, 43]

Major nutrient elements (symbol)	Content
Nitrogen (N)	2–5%
Phosphorous (P)	0.5–1%
Potassium (K)	1–2%
Calcium (Ca)	0.5–3%
Zinc (Zn)	15–60%
Copper (Cu)	4–20%
Sulphur (S)	0.2–3%
Magnesium (Mg)	0.3–1%
Iron (Fe)	500–1200 ppm
Manganese (Mn)	20–60 ppm

such as aphids, whiteflies, psyllids, leafminers, etc., while being safe for beneficial insects and having a low environmental impact. Despite initial manufacturing and regulatory hurdles, azadirachtin-based pesticides have become a key component in crop protection since their introduction in the US in 1993. However, they still represent a small fraction of the total insecticide market [37]. Neem oil has also been used as a biopesticide in crop protection for a long time, making pests somewhat resistant to Neem oil. Recent advances in nanotechnology can solve this problem. Nanoencapsulation of Neem oil could lead to more stable formulations for managing crop-damaging insects, particularly those vital for human nutrition. Furthermore, nanotechnology can help prevent insect resistance, often resulting from the widespread and untargeted application of Neem oil [38]. Neem cake (Fig. 4) is also noted for its efficacy in controlling insects and pests. Its bitter component has been documented to serve as an antifeedant, attractant, repellent, insecticide, nematicide, disruptor of growth, and antimicrobial agent [18]. Besides, Neem-coated urea can help to manage a variety of pests, including caterpillars, beetles, leafhoppers, borers, and mites. The properties of Neem-coated urea include antifeedant effects and pest growth regulation [39].

8 Neem as an organic fertilizer

The Neem tree is hailed as one of the most promising trees for fostering sustainable food production in the twenty-first century, according to the WHO/UNEP [20]. Neem products are now used as an indispensable agent in organic farming [40]. Neem cake, a by-product of the industrial extraction of Neem oil, contains a substantial nitrogen content ranging from 2 to 5% (refer to Table 4), which presents challenges for its utilization as an affordable fertilizer. Additionally, Neem cake is abundant in sulphur compounds and limonoids [41]. Table 4 documents the major nutrient elements present in Neem cake.

The US Environmental Protection Agency certified Neem-based products' effectiveness and environmental safety in 2012 [44]. Furthermore, under the European Parliament's 2019/1009 Regulation, Neem cake is recognized for its potential to enhance the cultivation of high-quality fruits and vegetables in an environmentally sustainable manner while maintaining the natural balance of the soil through its bio-stimulant properties [45]. In this regard, Neem cake's function as a bio-stimulant is linked to its role in sustainable nematode management, either by mitigating the impact of nematodes

on plant growth or by boosting plant resistance by providing essential nutrients for healthy plant development [46]. The versatile nature of Neem cake, serving as both a fertilizer and a soil conditioner with nematicidal and insecticidal properties, could significantly reduce the reliance on mineral fertilizers and pesticides. This, in turn, may lead to less mechanical soil disruption, lower environmental pollution, and a decreased dependence on fossil fuels [47, 48].

Neem cake enhances the organic content of the soil by lowering alkaline levels and generating organic acids as it decomposes. Since Neem is entirely natural, it harmonizes with soil microbes, enriches the rhizosphere microflora, and contributes to a more robust soil structure, better water retention, and improved aeration, supporting healthier root growth. Additionally, Neem cake shields plant roots from nematodes, soil grubs, and white ants. Neem contributes to soil fertility by containing a substance that prevents soil bacteria from converting nitrogenous compounds into nitrogen gas. As a nitrification inhibitor, Neem extends nitrogen availability for both short-term and long-term crops. When used alongside nitrogenous fertilizers, Neem can decelerate the transformation of nitrogenous compounds into ammonia, nitrates, and nitrites, enhancing overall soil efficiency. Neem cake is typically applied by incorporating it into the soil around and on the roots of plants [49–51]. Table 5 classifies the quantity of Neem cake required for different crops.

Lately, not only Neem cake but also Neem and its derivatives have been utilized to produce a urea-coating agent that enhances and sustains the efficiency of nitrogenous fertilizers in the soil. Due to its high solubility in water, urea's effectiveness is often diminished through nitrification and denitrification processes. By coating urea with Neem, these losses can be minimized, allowing for a gradual release of the fertilizer and extending its availability to plants over a longer period. Neem-coated urea is environmentally benign, as it reduces urea solubility and helps prevent groundwater contamination. The coating also decreases ammonia loss through volatilization, thereby mitigating atmospheric pollution.

Table 5 Dosage of Neem cake for different crops [42]

Name of the crop(s)	Quantity of Neem cake required*
Field crops	
Rice	500 kg/Ha
Wheat	600 kg/Ha
Sugarcane	1500 kg/Ha
Soyabean	1000 kg/Ha
Tobacco	750 kg/Ha
Potato	1000 kg/Ha
Vegetables/horticultural crops/trees	
Vegetables	1000 kg/Ha
Banana	300 g/plant
Coconut	3 kg/tree
Fruit trees	1 kg/tree
Lawn	100 g/sq. m
Flowering plants	
Rose	50 g/plant
Hibiscus	100 g/plant
Crotons	100 g/plant
Cut flowers	100 g/plant
Potted plants	
Small plant	50 g/pot
Large plant	10 g/plant
Plantation crops	
Coffee	1500 kg/Ha
Tea	1500 kg/Ha
Rubber	2 kg/tree
Oil palm	3 kg/tree
Cardamom	250 g/plant

* g: gram; kg: kilogram; Ha: Hectare; sq. m.: square meter

This eco-friendly practice benefits the environment and holds economic potential for farmers. The urea coating with Neem is typically applied in liquid or powdered form [52].

9 Use of Neem in cosmetics, topical products and dermatological care

Neem (*A. indica*) is widely used in cosmetics, topical products, and dermatological care due to its antimicrobial, anti-inflammatory, and skin-soothing properties [20]. Neem is a widely used ingredient in natural and Ayurvedic cosmetics, known for its versatile benefits in skin care and hair care. In facial creams and lotions, Neem oil and extracts are included for their antibacterial, antifungal, and antimicrobial properties, which help soothe irritated skin. In anti-ageing products, Neem's antioxidants, such as carotenoids, protect the skin from free radical damage, helping to prevent signs of ageing. It is also a common ingredient in facial cleansers and toners, where its antibacterial qualities help reduce acne and blemishes [53–55]. For hair care, Neem is incorporated into shampoos and conditioners to treat dandruff and scalp infections and encourage healthy hair growth [56, 57]. In topical products, Neem oil is applied directly to the skin to treat conditions like eczema, psoriasis, and acne due to its anti-inflammatory and antiseptic effects. It is also found in ointments and creams for managing fungal infections, wounds, and ulcers, as it supports healing and prevents secondary infections. Lip balms often include Neem extract to repair dry and cracked lips [19, 58]. Neem is also recognized for its dermatological benefits. It effectively treats acne by reducing sebum production and eliminating acne-causing bacteria, preventing future breakouts. Its moisturizing, anti-inflammatory, and antifungal properties make it valuable in managing chronic skin conditions such as eczema and psoriasis, as it soothes itching, reduces inflammation, and keeps the skin hydrated [59, 60]. Neem also aids in faster wound healing by promoting collagen production and lowering the risk of infection [61]. Its antifungal qualities are beneficial for treating infections like athlete's foot, ringworm, and candidiasis. Additionally, Neem's anti-inflammatory properties help reduce redness, swelling, and irritation from allergic reactions, dermatitis, or skin injuries [62, 63]. Neem is often combined with other natural ingredients like turmeric, sandalwood, and aloe vera in traditional and modern skincare formulations [64].

10 Use of Neem in ASU&H formulations

ASU&H formulations refer to preparations used in Ayurveda, Siddha, Unani, and Homeopathy, prepared using plants, minerals, metals, and products derived from animals and marine sources. Due to its extensive medicinal applications, the Neem tree is known as 'the village pharmacy' in India. Various parts of the tree, such as the bark, leaves, flowers, fruits, seeds, and oil, are utilized for diverse therapeutic purposes. Neem is a key component in many polyherbal formulations in the traditional system of Indian medicine and is also widely used by practitioners of Ayurveda, Siddha, Unani, and Homeopathy. Below, we have documented various ASU&H formulations containing Neem as an ingredient, along with their therapeutic effects.

In Sanskrit, the Neem tree is called '*Arista*', meaning 'reliever of sickness', and hence, in our Indian tradition, very often, it is called '*sarba roganibarini*', meaning the one who or which can treat all kinds of ailments [65, 66]. This is not an exaggeration as it is a panacea also in Ayurveda. In Ayurveda, Siddha, Unani and Homeopathy, i.e., in the ASU&H system, hundreds of formulations use Neem as an ingredient. Table 6 documents some important formulations where Neem and its components are used.

11 Responsible phytochemicals for bio-activities of Neem

The Neem tree is often called a "reservoir" of plenty of phytochemicals, with over 300 extracted and identified [74]. The isolated phytocompounds can be categorised primarily into two groups, i.e., isoprenoids, which include diterpenoids, triterpenoids, and limonoids, and non-isoprenoids, which contain proteins, polysaccharides, tannins, flavonoids, glycol-flavonoids, dihydrochalcone, coumarin, various aliphatic compounds, and phenolic acids [75, 76]. Every component of the Neem tree, including its leaves, fruits, seeds, oil, roots, bark, and twigs, is a treasure of phytochemicals known for their antimicrobial, antiviral, antifungal, anti-inflammatory, anti-plasmodial, antiseptic, antipyretic, anti-diabetic, and anti-ulcer activities [32]. Nimbin was the first compound to be explored. Other significant phytochemicals obtained from Neem include nimbolide, azadirachtin, azadiradione, gedunin, and azadirone.

Table 6 List of some ASU&H formulations containing Neem

Sl. No	Formulation(s) Name	Therapeutic uses
Ayurvedic formulation(s)		
Neem leaf (Lf.) as ingredient		
01	Panca Guna Taila [67]	Rheumatoid arthritis, Ear ache, chronic wounds
02	Manikya Rasa [67]	Skin diseases, skin inflammation, dermatitis, anaemia, jaundice
03	Arogyavardhini Gutika [67]	Skin diseases, hepatic disorder, chronic fever, improved digestion and food absorption
04	Jatyadi Ghrita [68]	Skin ulcer, piles, anal fistula
05	Jatyadi Taila [68]	Skin inflammation and redness, sinus, abscess, acts as antiseptic and fungicidal oil
06	Atasyadi Lepa [69]	Piles, fissures
07	Krmikantaka Rasa [69]	Anthelmintic activity, abdominal discomfort
08	Silajatvadi Vati [69]	Diabetes, boosts energy levels and immunity, and urinary tract-related issues
Neem Seed (Sd.) as ingredient		
01	Arosoghni Vati [67]	Piles, anal fissures, haemorrhoids
02	Guggulvadi Vati [69]	Joint pain (Arthritis, gout), detoxification
03	Madhumehari Churna [69]	Diabetes, Glycosurea, Albuminurea
04	Maharudraguduchi Taila [69]	Joint and muscle pain, skin diseases
Neem Seed oil (Sd. Oil) as an ingredient		
01	Manahsiladi Taila [69]	Skin disorders (prevents scar in the skin)
Neem Endosperm (Enm.) as an ingredient		
01	Muktadi Mahanjana [68]	Eye diseases
02	Arsoghni Vati [69]	Piles, haemorrhoids
03	Jvaramatangakesari Rasa [69]	Fever, body aches, weakness, fatigue
Neem *Panchang* (Stem bark, flower, leaf, root and fruit) as ingredients		
01	Pancanimba Churna [67]	Major and minor skin diseases, hyperacidity
Neem Stem Bark (St. Bk.) as an ingredient		
01	Nimbadi Kvatha Churna [67]	Skin diseases, detoxification, gout, arthritis
02	Pathayadi Kvatha (Sadanga) Churna [67]	Arthritis, joint pain
03	Brhanmaricadya Taila [67]	Skin diseases including psoriasis, eczema etc
04	Madhyama Narayana Taila [67]	Arthritis, cervical spondylosis, gout
05	Aragvadhadi Kvatha Churna [68]	Diabetes, skin diseases
06	Trayantyadi Kvatha Churna [68]	Skin diseases, hepatic disorders
07	Nimbadi Kvatha Churna [68]	Fever related illness
08	Punarnavastaka Kvatha Churna [68]	Oedema, Renal and hepatic disorders
09	Brhanmanjisthadi Kvatha Churna [68]	Skin disorder, blood purification

Table 6 (continued)

Sl. No	Formulation(s) Name	Therapeutic uses
10	Vasaguducyadi Kvatha Churna [68]	Jaundice, anaemia
11	Tiktaka Ghrita [68]	Skin inflammation, anaemia, thyroid dysfunction, hepatic disorder
12	Pancatikta Ghrita [68]	Psoriasis, weak bone, bone degeneration
13	Pancatiktaguggulu Ghrita [68]	Skin diseases, joint disorders, detoxification
14	Patoladi Ghrita [68]	ENT and eye disorders (cataracts, night blindness etc.)
15	Nimbadi Churna [68]	Blood purification and detoxification, digestive disorders
16	Sudarsana Churna [68]	Chronic fever
17	Maha Visagarbha Taila [68]	Arthritis, joint and muscular pains, hemiplegia
18	Vajraka Taila [68]	Rheumatoid arthritis, non-healing wounds secreting pus
19	Siva Gutika (Laghu) [68]	Skin diseases, anaemia, piles, diabetes
20	Kalameghasava [69]	Liver disorders, including cirrhosis, splenomegaly, constipation
21	Candanadyarka [69]	Gastritis, urinary infections, skin disorders
22	Amrtastaka Kvatha Churna [69]	Digestive and respiratory health, immune boosting property
23	Navakarsika Kvatha Churna [69]	Skin diseases, gout
24	Phalatrikadi Kvatha Churna [69]	Digestion-related problems, constipation
25	Laghu Manjisthadi Kvatha Churna [69]	Skin disorders, blood purification, digestive health
26	Haridradi Kvatha Churna [69]	Hepatoprotective, skin and digestive disorders
27	Varadi Guggulu [69]	Obesity, Boosts metabolism and digestive health
28	Sodasanga Churna [69]	Indigestion, boosts digestive health, detoxification
29	Jvarasamhara Rasa [69]	Chronic fever
30	Astadasanga Lauha [69]	Anaemia, improves digestion
31	Yakrdari Lauha [69]	Liver disorders, splenomegaly
Siddha Formulation(s)		
01	Visnucakkara Mattirai [70] (Neem Fruit is used)	Diabetes and metabolic disorders
02	Cinthil Ney [70] (Cinthil oil) (Neem Stem Bark is used)	Skin-related disorders, skin inflammation and moisturization of the skin
03	Cirakac Curaman [70] (Neem Stem Bark is used)	Skin-related disorders, minor wounds and skin inflammation
04	Mandurati Adaikkudinir [70] (Neem Stem Bark is used)	Anaemia, blood purification, and overall nourishment of the body
05	Tampiraccenturam [70] (Neem Flower and Flower bud are used)	Respiratory problems and skin inflammation
Unani formulation(s)		
01	Majoon Musaffi-e-Khoon [71] (Neem Root is used)	Purification and detoxification of blood
02	Habb-e-Surkhbada [72] (Neem Leaf is used)	For treatment of erysipelas (a bacterial skin infection) and skin inflammation
03	Habb-e-Siyah Chashm [72] (Tender Neem Leaf is used)	For treating conjunctivitis, itchy and dry eyes
04	Marham-e-Jadwar [72] (Tender Neem Leaf is used)	For treating ulcers, lymphadenitis etc
05	Araq-e-Juzam [72] (Neem Whole Plant is used)	Purification of blood and detoxification in general, leprosy, leucoderma, scabies etc
06	Habb-e-Neeb [72] (Neem Seed is used)	For treating respiratory problems, acne, pimples etc

Table 6 (continued)

Sl. No	Formulation(s) Name	Therapeutic uses
07	Araq-e-Musaffi-Ekhoon Qawl [72] (Neem Leaf is used)	For treatment of chronic skin diseases like psoriasis, dermatitis, purification of blood etc
08	Habb-e-Bawasir [72] (Neem Stem Bark is used)	For treating skin disorders, piles, detoxification of blood etc
Homeopathy formulation(s)		
01	Azadirachta Indica Mother Tincture Q [73] (Neem Bark is used)	For treating skin diseases, maintaining hair and scalp health, oral health, digestive health and boosting immunity

[#] In the above-mentioned ASU&H formulations, Neem plant or its parts are one of the ingredients, but the formulation may contain many other ingredients in the form of other plants, minerals, metals, and products derived from animals and marine sources. So, medication must be administered under the supervision of a qualified Ayurvedic/Siddha/Unani/Homeopathic practitioner due to the presence of potentially toxic minerals and herbs, which require proper processing and dosage

Azadirachtin comprises seven isomeric chemical entities, known as azadirachtin A-G, with azadirachtin E being the major. The most biologically potent triterpenoid, Nimbin, is found in Neem seed oil and is recognised for its antipyretic, antiseptic, anti-inflammatory, fungicidal, and antihistamine effects [77, 78]. Neem Leaf Glycoprotein (NLGP), derived from Neem leaves, is a purified protein known to decelerate tumour cell proliferation. Rather than directly killing cancerous cells, this protein alters immune cell behaviour within the cancerous environment and the bloodstream, effectively hindering cancer progression [79]. Table 7 lists some essential phytochemicals in the Neem plant and their bio-activities.

12 Safety and toxicities associated with Neem

Evaluating the toxicities of natural compounds is critical before their use in health management. Various animal model studies and clinical trials have confirmed that Neem is safe at specific doses. However, Neem and its components have also demonstrated toxic or adverse effects. Regular use of neem extracts or compounds has been reported to pose safety concerns, potentially leading to toxicity and, in severe cases, even lethality [100]. Several studies have reported that Neem oil poisoning in children can lead to symptoms such as vomiting, hepatic toxicity, metabolic acidosis, and encephalopathy [15, 101]. Another study using a rat model found that administering Neem leaf sap produced an antianxiety effect at low doses, but this effect was not observed at higher doses [102]. A critical study involving rats revealed that azadirachtin did not exhibit toxicity even at a dose of 5 g/kg body weight [103]. Furthermore, a study conducted on rabbits to assess the toxicological impact indicated a steady increase in body weight in both the test and control animals, with no signs of toxicity observed in either group throughout Neem extract administration [104]. A study determined the LD50 for Neem leaf and stem bark extracts to be 31.62 mg/kg and 489.90 mg/kg body weight, respectively [105]. For aqueous extracts of *A. indica* leaves and seeds, the LD50 values were found to be 6.2 mL/kg and 9.4 mL/kg, respectively [106]. An acute oral toxicity test in mice established an LD50 value of approximately 13 g/kg body weight [107]. NIM-76, a volatile neem oil fraction, exhibits antifertility effects when applied before coitus in rats, rabbits, and rhesus monkeys [108–110]. Nimbidin has been shown to cause subacute toxicity in adult rats following daily doses of 25, 50, or 100 mg/kg for 6 weeks. Its hypo-glycaemic effect has also been observed in fasting rabbits and exhibits spermicidal activity as well. Nimbolide and nimbic acid are toxic to mice when administered intravenously or intraperitoneally but are less harmful to rats and hamsters. Higher doses of these compounds can result in death due to multiorgan failure, and a sudden drop in arterial blood pressure is observed [111]. Nimbolide has demonstrated cytotoxic effects on NIE-115 neuroblastoma (mouse), 143B TK-osteosarcoma (human), and Sf9 (insect) cultured cell lines, with IC30 values of 27 and 112 µm, respectively [112, 113]. Nimbidin, deacetyl nimbin, and azadirachtin are considered nontoxic. Studies on Margosan O, a neem seed extract, reveal no apparent toxicity in mammals but indicate some toxicity in rats and mice. Acute dermal toxicity has been reported at 2 mg/kg in albino rabbits, though it causes minimal eye irritation [114–116].

13 Pharmacological studies carried out on Neem

Over 300 phytocompounds have been extracted and identified from various parts of Neem, which have plenty of therapeutic benefits [74, 117]. The role of these phytocompounds is well established, as evidenced by multiple research papers. In Ayurveda, the role of a single phyto-compound is not considered; instead, the synergistic effect of numerous phytochemicals present in herbal extract is considered. So, below is a list (Table 8) of pharmacological studies on herbal extracts of various parts of Neem.

14 Therapeutic effects of Neem

Since ancient times, humans have widely utilized different parts of the Neem, including bark, flowers, leaves, seeds, and twigs, to treat various ailments. The following table (Table 9) documents the therapeutic effects of the Neem and its various parts.

The main intention of the inclusion of Table 9 was to provide a summary of the therapeutic effects of Neem and its various parts. There are two options: either to consume the Neem as a *vati* (tablet), *churna*, etc. or to apply it topically in

Table 7 Bio-activity of phytocompounds extracted from Neem plant

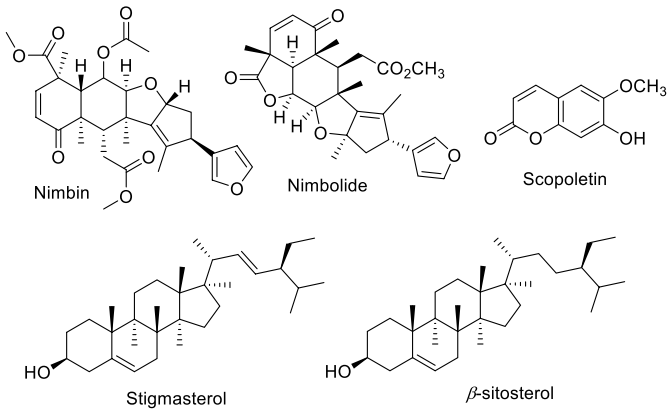
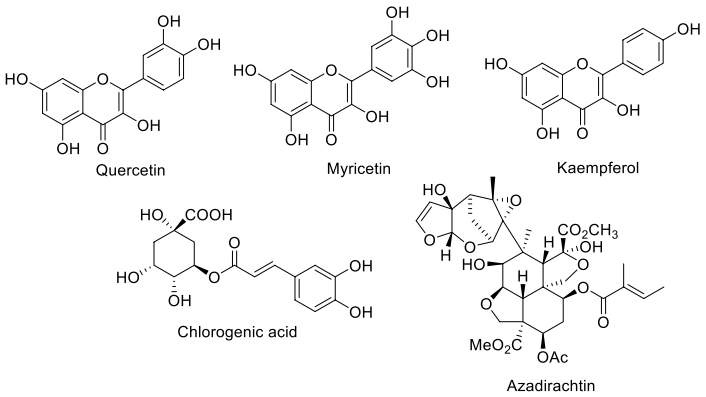
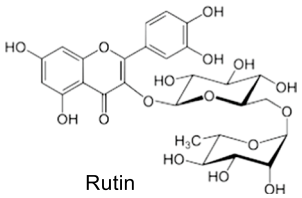
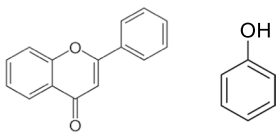
Plant part	Bio-activity	Responsible phytochemicals
Leaf [80]	Anti-oxidant	
	Anti-inflammatory	
	Anti-hyperglycaemic	
Bark [81]	Anti-oxidant	

Table 7 (continued)

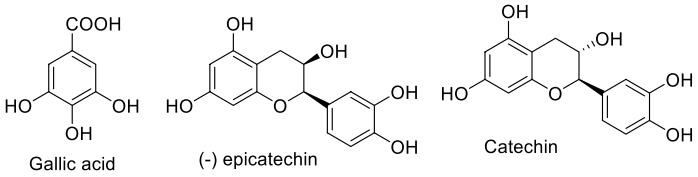
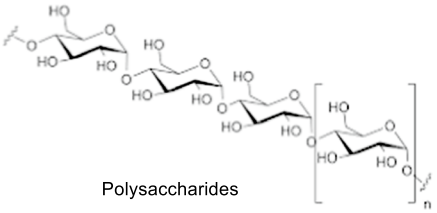
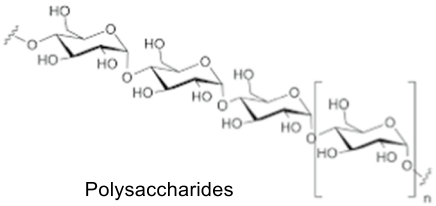
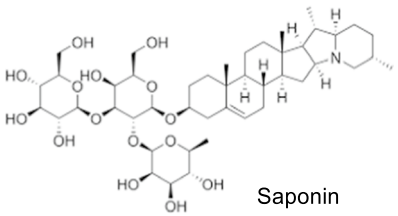
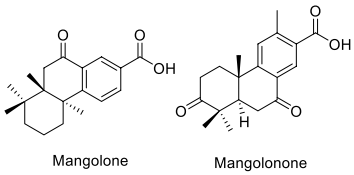
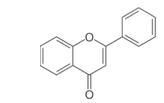
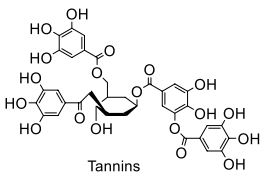
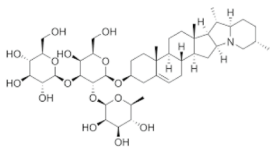
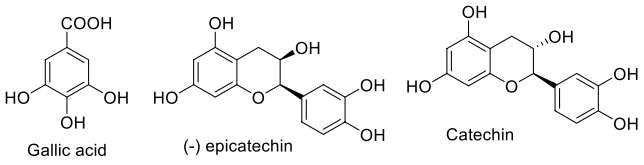
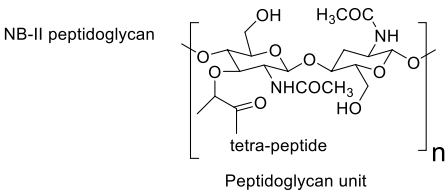
Plant part	Bio-activity	Responsible phytochemicals
	Anti-inflammatory	 
	Anti-tumour	
	Haemolytic	
	Antimicrobial	   
	Immunomodulator	 

Table 7 (continued)

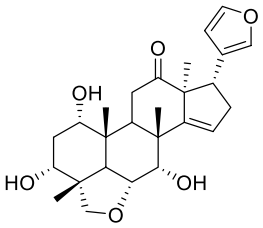
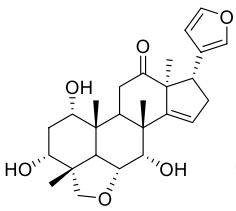
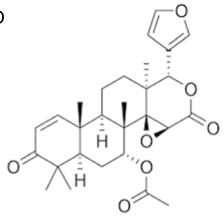
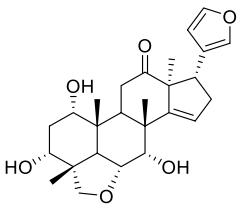
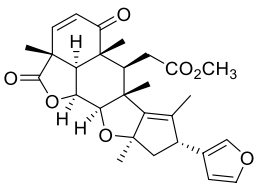
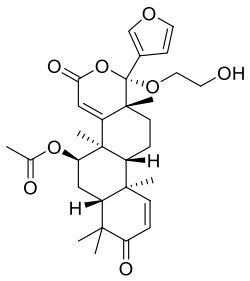
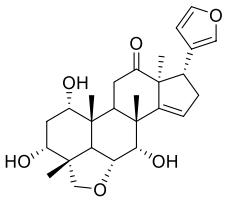
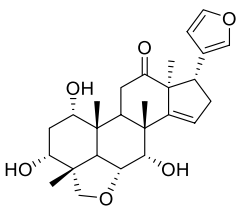
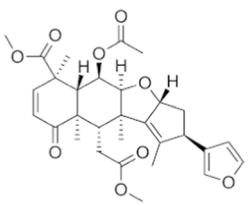
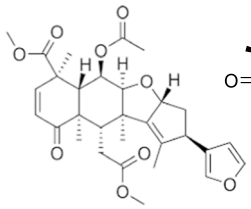
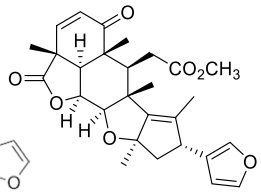
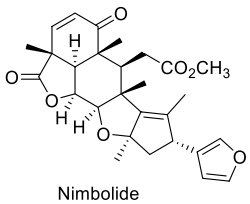
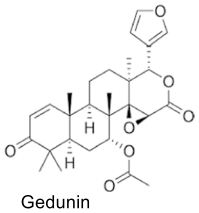
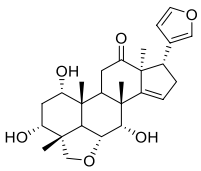
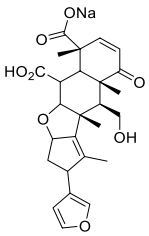
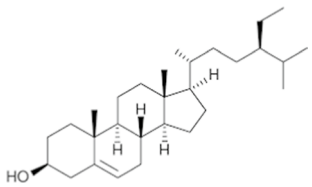
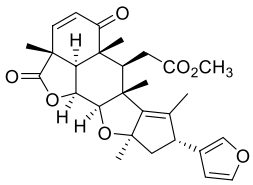
Plant part	Bio-activity	Responsible phytochemicals
Seed [82]	Hypoglycaemic	 Nimbidin
	Anti-arthritic	
	Anti-gastric	
	Anti-fungal	 Nimbidin  Gedunin
	Anti-bacterial	 Nimbidin  Nimbolide  Mahmoodin
	Diuretic	 Nimbidin
	Antipyretic Spermicidal	 Nimbidin  Nimbin

Table 7 (continued)

Plant part	Bio-activity	Responsible phytochemicals
	Antioxidant	 Nimbin  Nimbolide
	Anti-malarial	 Nimbolide  Gedunin
	Anti-inflammatory	 Nimbidin  Sodium nimbidate
Fruit [83]	Anti-ulcer	 Phytosterol (representative structure)
Flower [84]	Anti-arthritic	 Nimbolide

* The list is not exclusive; only important compounds in terms of their therapeutic properties are listed in this Table. Plenty of compounds are present in the Neem, and plenty of compounds have been extracted from different parts of the neem tree, though only a few have undergone biological activity studies [85, 86]. Nimbidin, a bitter compound extracted from the seed kernel oil of Neem, is known for diverse biological activities. Neem oil contains several tetra-nor triterpenes, such as nimbin, nimbidin, nimbolide, and nimbidic acid, which have been identified [87]. Both nimbidin and sodium nimbidate exhibit significant dose-dependent anti-inflammatory effects, reducing carrageenan-induced acute paw edema in rats and formalin-induced arthritis. Nimbidin has also been confirmed to possess antipyretic properties [88, 89]. When administered orally, it significantly lowers blood glucose levels in fasting rabbits [82]. Furthermore, nimbidin shows a marked antiulcer effect, preventing gastric lesions induced by acetylsalicylic acid, indomethacin, stress, or serotonin, as well as duodenal ulcers caused by histamine or cysteamine [90]. It suppresses basal and stimulated gastric acid secretion and may function as an H_2 receptor blocker, contributing to its antiulcer potential. Nimbin and Nimbidin have also demonstrated spermicidal effects in rats and humans [91, 92]. Nimbidin possesses antifungal properties, inhibiting the growth of *Tinea rubrum*, and can completely suppress *Mycobacterium tuberculosis* in vitro, exhibiting bactericidal effects [93, 94]. Azadirachtin, a highly oxygenated C-secomeliacin from neem seeds, is known for its antifeedant properties and has shown antimalarial activity. Sodium nimbidinate has been reported to have diuretic effects in dogs. At the same time, nimbolide inhibits the growth of *Plasmodium falciparum* and displays antibacterial activity against *Staphylococcus aureus* and *S. coagulase* [95–98]. Gedunin, extracted from neem seed oil, exhibits antifungal and antimalarial properties, inhibiting malarial parasite development [96]. Mahmoodin, a deoxy gedunin, from neem seed oil, demonstrates moderate antibacterial activity against various human pathogens [91]. Additionally, condensed tannins from neem bark contain gallic acid, (+)-gallocatechin, (−)-epicatechin, (+)-catechin, and epigallocatechin. Gallic acid, (−)-epicatechin, and catechin, in particular, are responsible for inhibiting chemiluminescence generation by activated polymorphonuclear neutrophils (PMN), suggesting their role in suppressing the oxidative burst associated with inflammation [99]

Table 8 Examples of a few pharmacological studies carried out on Neem

Pharmacological activity	Extract type	Organism(s)	References
Anti-oxidant	Aqueous, methanolic and ethanolic extract of Root and Bark	DPPH scavenging assay	[118]
	Aqueous leaf extracts	Phospho-molybdenum assay	[119]
	Neem crude oil extracted from the seeds	DPPH scavenging assay	[120]
	Ethanolic extract of flowers and seed oil	DPPH scavenging assay	[121]
Anti-nociceptive	Methanolic Extract of Leaves	Heat-induced-mechanical and chemical-induced Nociception models in mice	[122]
	Ethanolic Extract of Fruit	Adult zebrafish (*Danio rerio*) model	[123]
Anti-inflammatory	Methanolic Extract of Leaves	Human leukaemia cancer cell lines	[124]
	Polysaccharides extracted from seeds	Wistar rats (Models of acute inflammation)	[125]
	Neem seed oil	Albino rats (Models of carrageenan-induced hind paw edema)	[126]
Anti-hypertensive	Methanolic Extract of Leaves	Wistar rats	[127]
Anti-hyperglycaemic	Neem leaves powder	Albino rats	[128]
	Aqueous extract of leaves and twigs	alloxan-induced diabetic rats	[129]
Anti-tumour	Ethanolic Extract of Leaves	Breast Cancer Cells of BALB/c Mice	[130]
	Ethanolic Extract of Leaves	OSCC human cell lines	[131]
	Supercritical CO_2 extract from Neem leaves	Human colorectal cancer cell lines (HCT116 and HT29)	[132]
Haemolytic	Mexico Neem leaf tea	A clinical study on a 35-year-old Hispanic man having deficiency of G6PD (glucose-6-phosphate dehydrogenase)	[133]
Anti-microbial	Aqueous extracts of leaf, bark and seeds	Human pathogenic bacteria (*S. aureus*, *E. feacalis*, *P. mirabilis* and *P. aeuroginosa*) and fungi (*A. fumigatus* and *C. albicans*)	[134]
	Methanolic extract of Neem plant	In vitro dentin disinfection model (*S. mutans*, *E. faecalis*, *S. aureus* and *C. albicans*)	[135]
Anti-fungal	NIM-76 (a volatile fraction of Neem oil)	*E. coli* and *P. aeruginosa*	[136]
Anti-viral		poliovirus (vaccine strain) in vero cell line	
Immunomodulator		Rats (In vivo)	[137]
	Methanolic extract from leaves	Mice (*Mus musculus*) (In vivo)	[138]
Anti-arthritic	Nimbolide (isolated from the Neem leaf and flower)	Freund's adjuvant induced arthritis in rats (in vivo)	[84]
	Nimbolide (isolated from Neem seed oil)		[84]
	Nimbidin (isolated from Neem seed oil)	Rats (in vivo)	[139]
Hepatoprotective activity	Alcoholic extracts of Neem leaf	Rat (in vivo) (paracetamol-induced liver damage)	[140]
	Aqueous extracts of Neem leaf		[109]
	Ethanolic extract of Neem leaf	Sprague Dawley adult male rats (in vivo) (paracetamol-induced liver damage)	[141]
Anxiolytic and antidepressant-like activities (neuropharmacological activity)	Aqueous extract of Neem flower	Male Wistar rats (in vivo)	[142]
	Acetone extract from Neem leaf	Rats (in vivo)	[102]

Table 8 (continued)

Pharmacological activity	Extract type	Organism(s)	References
Antipyretic	Chloroform extract from leaves	albino rats (in vivo)	[143]
	Neem leaf extract	albino rats (Brewer's yeast-induced pyrexia model) (in vivo)	[144]
Spermicidal	Aqueous extract of Neem leaf (fresh and old)	Human sperm (in vitro)	[145]
	Extract from Neem seed		[146]
Anti-malarial	Methanolic extract of Neem leaf and stem bark	Albino Mice Infected with *Plasmodium berghei* (in vivo)	[105]
	Ethanolic Extract of Neem Leaf		[147]
	Ethanolic Extract of Neem Leaf	C57BL mice infected with *Plasmodium berghei* (in vivo)	[148]
Anti-ulcer	Aqueous extract of Neem leaves	Wistar rats (Pyloric Ligation-Induced ulcer model, Aspirin-Induced Gastric ulcer model, Cold Restraint Stress-Induced ulcer model)	[149]
	Ethanolic extract of Neem stem bark	Albino rats	[150]

Table 9 Various parts of the Neem tree and their therapeutic effects

Plant part	Therapeutic effects
Whole plant [77]	All parts of the Neem tree, i.e., stem and root bark, flowers, leaves, fruits, and seeds, have been traditionally used to treat inflammation, infections, fever, skin diseases and dental disorders
Bark [151]	Analgesic, stomach and intestinal ulcers, skin diseases, fever, etc
Leaf [152]	Skin diseases, eye problems, bloody nose, diseases of the heart and blood, skin ulcers, diabetes, gum disease, useful for birth control and to cause abortions, etc
Flower [153]	Bile suppression, elimination of intestinal worms, etc
Fruit [154]	Piles, urinary tract disorders, mouth ulcers, bloody noses, hair problems, and to kill lice, pimples, cholera, phlegm, piles, etc
Seed (seed oil) [155]	Intestinal worms, skin diseases etc
Twig [156]	Cough, asthma, as a toothbrush to clean teeth etc
Gum [157]	The older trees excrete sap, which is used to treat skin diseases and provide oral health care

the form of lotion, cream, balm, etc. The qualified practitioner prescribes the medicine with the proper dosage depending on the indications. In the traditional system of Indian medicines, the Neem plant holds an important position for its versatility. Table 10 showcases some of its therapeutic effects as per the Indian system of traditional medicine.

15 Patent information on various neem formulations

In this part of our study, we utilized freely accessible databases, like Patent Scope-WIPO, Espacenet and Google Patents, to conduct a patent search. We employed the search with the keyword '*Azadirachta indica*' in the search button. Following is the list of some patents:

IN202011023472:	The preparation process of an herbal face mask using *Azadirachta indica* and *Saraca asoca* leaf is described [162]
CN109805037A:	Hexane extract of Neem seed acts as a bacteria-killing agent [163]
KR1020200102018:	Neem leaf extract (hot water/ethanol/acetone) is helpful for acne alleviation and amelioration of whitening and wrinkling [164]
MX2013004928A:	Neem leaf extract is used for skin rejuvenation and treatment of fungi and bacteria [165]
IN2177/CHE/2010:	Neem leaf is used to prepare biogenic nanoparticles used for biomedical applications [166]
WO2020020539A1:	Aqueous Neem extract is used as an anti-dandruff shampoo [167]
IN202311006650:	A gel formulation containing the stem bark of Neem and has the potential of wound healing [168]
CN108434032A:	Neem leaf extract is used as a skin care product for removing acne [169]
IN617/KOL/2007:	Aqueous extract of bark and root for diabetes treatment [170]
CN108158972A:	Neem leaf extract is used to inhibit acne inflammation [171]
WO/2020/047405:	Describes preparation method of supercritical CO_2 neem leaf extract to treat oral and colon cancer in patients [172]
NZ578363A:	Formulation contains aqueous extract of Neem bark for topical wound healing [173]

16 Neem-based clinical studies

Numerous clinical trial-based studies have confirmed that herbal products and derivatives from natural sources play a crucial role in preventing and treating diseases. However, there have been only a few studies on active compounds like nimbidin to assess their efficacy in health management. One significant study involved human subjects to evaluate Neem bark extract's antisecretory and antiulcer effects. The administration of lyophilised extract powder at 30 mg twice daily for 10 days resulted in a substantial reduction (77%) in gastric acid secretion. Further, administering the bark extract at 30–60 mg twice daily for 10 weeks almost completely healed duodenal ulcers. In one case, each

Table 10 Therapeutic effects/pharmacological activities of Neem in ASU&H (Ayurveda, Siddha, Unani and Homeopathy) system

Ayurveda	Neem, a widely used plant in Ayurvedic medicine, serves various purposes in treating skin disorders and diabetes. Here are some common home remedies using Neem [158]: Skin Disease: 10 ml of Neem leaf juice with honey (twice daily) and bathing with water boiled with Neem leaves Wounds/Ulcers: Applying Neem leaf paste on affected areas alleviates wounds and ulcers Intestinal Issues: Drinking a decoction from a handful of Neem leaves (20 ml) on an empty stomach for three (03) days helps with intestinal wounds Loss of appetite: Decoction made from a handful of leaves (20 mL) should be taken on an empty stomach for three (03) days Dandruff/Ringworm: Applying a decoction of Neem leaves to the scalp and affected areas 1 h before bathing can heal dandruff and ringworm Diabetes: Consuming 5 gm of dried Neem leaves/fruit powder with lukewarm water on an empty stomach (twice daily) is beneficial for some instances of non-insulin-dependent diabetes mellitus in its primary stages
Siddha	In Siddha medicine, Neem purifies the blood, alleviates inflammation, and aids in healing. It treats various skin ailments such as infections, eczema, psoriasis, scabies, and ringworm. Neem is often combined with other herbs like turmeric and sesame oil in formulations, mainly in paste or ointment form, prescribed by Siddha doctors, aiming to reduce inflammation, relieve itching, and facilitate skin recovery. Neem oil, derived from Neem leaves, is widely applied in Siddha medicine for its purported anti-inflammatory and antiseptic properties. It can be used topically or added to bath water to soothe skin irritation [159]
Unani	Neem also holds a prominent position in the Unani medicine system because of its many pharmacological activities (*AFAAL*). Following is a documentation of pharmacological activities mentioned in several classical Unani texts [160] • *Dafe humma* (Antipyretic) • *Dafe tafffun* (Antiseptic) • *Habise ishaal* (Antidiarrheal) • *Munzij* (Concoctive) • *Nafe amrazz jild* (Skin Diseases) • *Musakkine atash* (Thirst quenching) • *Muqavvi basar* (Eye tonic) • *Mudire Haiz* (Emmanagogue) • *Mane nazool* (Anti hydrotic) • *Nafe zeequnnafas* (Beneficial in asthma) • *Nafe wajaul mufasil* (Anti rheumatic) • *Nafe falij* and *laqwa* (Beneficial in paralysis and hemiplegia) • *Dafe yarqaan* (Beneficial in Jaundice) • *Hazim* (Digestive) • *Kasire riyah* (Anti flatulent) • *Musakkine alam* (Analgesic) • *Mundamile Qurooh* (Cicatrizant) • *Muqavvi snan wa lissa* (Teeth and gum tonic) • *Musakkine suaal* (Antitussive) • *Musakkine Suda* (Headache Reliever) • *Nafe ziabetus* (Anti Diabetic) • *Nafe istisqa* (Useful in Dropsy) • *Nafe darde gosh* (Useful in ear ache) • *Nafe jarooh wa qurooh* (Beneficial in burns and wounds)
Homeopathy	In homoeopathy medicine, Neem is believed to have various potential benefits. Following are some key benefits associated with Neem [73, 161]: Skin Conditions: It treats acne, eczema, psoriasis, and other inflammatory skin issues thanks to its antibacterial, antifungal, and anti-inflammatory properties Hair and Scalp Health: Helps manage dandruff, itchy scalp, and hair loss by promoting scalp health and addressing related conditions Oral Health: Addresses gum diseases, tooth decay, and bad breath with its antimicrobial properties, supporting oral hygiene Digestive Health: Aids in digestion, alleviating issues like indigestion, bloating, constipation, and regulating bowel movements Immune Support: Boosts the immune system during illness or periods of weakened immunity, believed to have immune-enhancing properties

oesophagal and gastric ulcer healed completely with 30 mg twice daily for 6 weeks [174]. A double-blind clinical trial assessed the efficacy of a drug made from an aqueous extract of Neem leaves in 50 cases of uncomplicated psoriasis alongside the conventional coal tar regimen. The results indicated that patients who received the drug in addition to

coal tar therapy showed a faster and better response compared to the placebo group [175]. In another clinical study lasting 6 weeks, the effectiveness of a Neem extract dental gel was compared to chlorhexidine gluconate (0.2% w/v) mouthwash as a positive control. The study found that the dental gel containing Neem extract significantly reduced plaque index and bacterial count compared to the control group [176]. Additionally, a study demonstrated that nimbidin significantly promoted ulcer healing in acetic acid-induced chronic gastric lesions in albino rats and dogs [177].

17 Synergy in the life cycle of Neem, plants, animals, and humans

The Neem tree (*A. indica*) is an extraordinary plant that engages with both the plant and animal kingdoms, as well as humans, in distinct ways throughout its life cycle. Below, we have discussed the interconnectedness between Neem, plants, animals, and humans throughout its life cycle.

The Neem tree begins its life in the plant kingdom right from the germination of seeds. It adapts to different soil types and can flourish even in dry environments. Neem contributes significantly to the plant ecosystem, with its deep roots preventing soil erosion and enhancing soil fertility by fixing nitrogen. In agroforestry, its leaves and seeds are natural pesticides and fertilizers, encouraging healthy plant growth without chemicals. Neem-based products, like Neem oil and Neem cake, possess pest-repellent qualities, making them valuable in organic farming for safeguarding crops from harmful pests, benefiting the overall plant ecosystem and promoting the sustainability of the environment as well [27, 178].

Neem trees provide shelter for birds, insects, and animals, serving as a habitat for many species, especially pollinators. Various animals feed on different parts of the tree; for instance, birds eat Neem fruits, and livestock may consume its leaves, though its bitterness makes it less favoured by grazing animals. Neem's natural compounds, such as azadirachtin, repel insects, safeguarding both crops and animals from pests without the use of synthetic pesticides. Insects like locusts avoid Neem-treated areas, minimizing harm to plant and animal habitats. Although Neem has insect-repellent qualities, its leaves and flowers still attract bees, butterflies, and beetles, with bees searching for nectar. Ants are attracted to the sap and insects found on Neem trees, forming mutualistic relationships by protecting the tree from herbivorous insects. As a result, Neem trees play a key role in supporting diverse wildlife and maintaining the ecological balance within their habitats. [38, 179].

Neem holds a prominent position in traditional medicine, widely used in the formulations of ASU&H drugs, with humans using its leaves, roots, seeds, fruits, flowers, and bark for medicinal, agricultural, and cosmetic purposes, fostering its widespread cultivation. Neem's leaves, bark, seeds, and oil are used to treat various conditions, from skin disorders and digestive problems to infections, with its medicinal use spanning from time immemorial. In addition to its healing properties, Neem is a natural pesticide commonly applied in organic farming to safeguard crops. Farmers rely on Neem oil and cakes to defend against various agricultural pests. Neem is also widely utilized in personal care items like soaps, toothpaste, and shampoos, valued for its antibacterial and anti-inflammatory benefits in hygiene and skincare. People use Neem wood to build furniture, kitchen accessories, etc., because of the strength and durability of Neem heartwood. Moreover, Neem has deep cultural and spiritual significance. For their protective and purifying qualities, Neem trees are planted near homes and temples in India and many South Asian regions [19, 82, 180–183].

When Neem trees die, their decomposition enriches the soil, providing a fertile environment for the next generation of plants and helping the decomposers (fungi, bacteria, etc.) to sustain in the animal kingdom. Their fallen leaves serve as natural fertilizer and mulch, further contributing to the life cycle of the plant kingdom.

Thus, the Neem tree completes its life cycle, right from its germination to death, by aiding, nourishing, interacting, enriching, and benefiting both the plant and animal kingdoms, as well as humans, at various stages.

18 Future perspectives

Future research on Neem (*Azadirachta indica*) in traditional systems of medicine offers immense potential for advancing therapeutic applications and enhancing evidence-based practices. Continued isolation and characterization of bioactive compounds, particularly limonoids, flavonoids, and glycoproteins, are essential to uncover new pharmacological applications. Standardization of ASU&H formulations, including extraction methods, dosage forms, and formulation protocols, is crucial to ensure efficacy and safety. Rigorous clinical trials are required to validate traditional

claims and establish the long-term safety and effectiveness of Neem-based treatments for dermatological, metabolic, and microbial conditions. Integrating traditional formulations with modern pharmacological agents can pave the way for innovative cosmeceuticals and nutraceuticals. In agriculture, sustainable cultivation techniques, including Neem's role as a bio-pesticide and soil conditioner, can contribute to environmental conservation. At the same time, advancements like nano-encapsulation may enhance the bioavailability and effectiveness of Neem products. Comparative studies exploring Neem's Siddha and Unani medicine applications for skin diseases and blood purification present valuable opportunities. Genomic studies can further identify genes responsible for bioactive compound synthesis, fostering genetic advancements for enhanced medicinal yields. Documenting and preserving indigenous knowledge, including folklore and ethnobotanical practices, can safeguard traditional wisdom and serve as a foundation for future scientific pursuits. Thus, we anticipate that this review will serve as a comprehensive resource for researchers, facilitating the integration of traditional knowledge with modern scientific advancements and paving the way for future studies to unlock Neem's full therapeutic potential, thereby ensuring its continued relevance and application in traditional systems of medicine.

19 Conclusion

The popularity of natural products and their derivatives in disease prevention and treatment is rising globally, largely due to the low risk of side effects. Neem and its components hold therapeutic significance and have been traditionally utilized for centuries. The review delves deeply into Neem's diverse therapeutic uses and cultural importance, emphasizing its historical and traditional role, particularly in the Indian subcontinent. Neem's significance is well-established across various traditional medicine systems, including Ayurveda, Siddha, Unani, and Homeopathy (ASU&H). Since ancient times, humans have utilised different parts of the Neem tree to treat various ailments. Recognized for its extensive medicinal utility, Neem treats various health issues, such as skin ailments, diabetes, and digestive disorders, utilizing different plant parts—leaves, bark, seeds, flowers, and oil. Modern science has discovered appropriate responsible molecules behind the numerous bio-activities of Neem and its different parts. The pharmacological effectiveness of Neem stems from its rich phytochemistry, which includes compounds like Nimbin, azadirachtin, nimbolide, stigmasterol, β-sitosterol, to name a few. These compounds contribute to Neem's antibacterial, anti-inflammatory, antifungal, antidiabetic, anti-ulcer, and immunomodulatory activities, making them effective for numerous health conditions. Clinical studies have verified Neem's crucial role in preventing various diseases. Beyond medicinal applications, Neem is also valued in organic farming and pest control, where Neem oil and Neem cake provide eco-friendly solutions. Recent advancements like nanoencapsulation are discussed, which could boost Neem's efficacy in agriculture by counteracting pest resistance. Culturally, Neem is revered in India, with associations to deities and applications in rituals, protective practices, and traditional cuisine. The review further highlights Neem's use in dermatology and cosmetics, where its skin-beneficial properties lend themselves to anti-ageing and antimicrobial applications. Overall, the review underscores Neem's role as a traditional therapeutic and sustainable resource, affirming its relevance for health and ecological management. This review article is anticipated to build a strong foundation for understanding the phytochemicals present, pharmacological activities, and medicinal applications of various components of the Neem tree, highlighting its role and applications in the ASU&H system.

20 Methods followed in this review

Relevant references concerning Neem (*Azadirachta indica* A. Juss) described in various ancient texts (*Samhitas* and *nighantus*, etc.) were reviewed. The author retrieved valuable publications from online databases like Web of Science, Science Direct, PubMed, Scopus, etc. Google search was also used. Online materials were also considered. All the references were collected and reviewed thoroughly.

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Data availability No datasets were generated or analysed during the current study.

Declarations

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